

QUANTUM COMPUTING

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Introduction

Grover's Algorithm

Solving Sudoku puzzle Qiskit





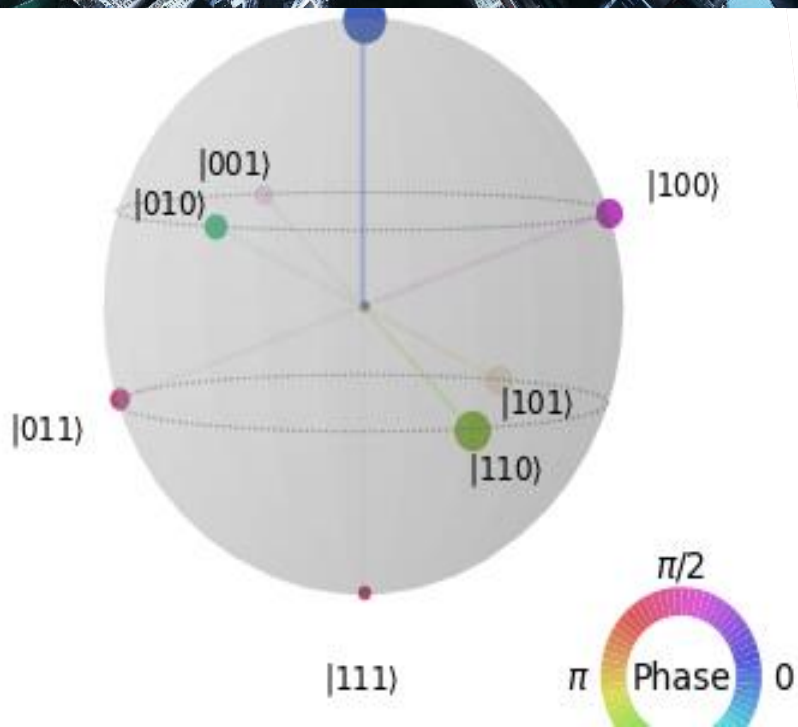
INTRODUCTION

Quantum Computing

Quantum computing is a fast-developing field and is promising to change how we approach computing. For some time, we are used to classical computers that is represented by classical states of “0” or “1”

Quantum computing now introduces possibilities of a state being in a super position. This has led to the development of some quantum algorithms that can speed up operations of some computations that had exponential growth with classical computing. The aspect of interference allow us to design quantum circuits so that we can amplify the desired solution and minimize the non-solutions.(Qiskit, 2020)

In this presentation we will look at one of such algorithms, specifically the Grover search algorithm.



Grover's Algorithm

- Quantum search algorithm
- Developed by Lov Grover in 1996
- Searches through unordered list faster than a classical algorithm. With exponential advantage
- If we have N number of entries. A classical algorithm would take N queries or $N/2$ on average to find the solution. Grover's algorithm works in $\sqrt{N}$
- We can also describe the Grover's algorithm as a function to find $f(x)$ such that $f(x)=1$



THE GROVERS ALGORITHM IN SUMMARY

Step 1: Initializing the system to a state. Create search space.

Step 2: Perform the Grover's Iteration $r(N)$ times. (Oracle and diffusion operator)

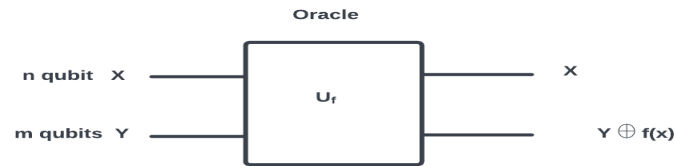
Oracle marks the correct answers. *An example we see later is the Soduku puzzle. We give the oracle the possible solution and the oracle can confirm if the answers are correct.*

Diffusion operator magnifies these answers so they can stand out and be measured.

Step 3: Perform the measurement.

THE ORACLE

- We assume that we have a device that we pass queries to and returns answers. Example we will look later is the Sudoku puzzle. We can give the oracle our solution and the oracle can easily check if it satisfies the condition and returns true or false.
- Unlike the classical algorithm where the functions are not reversible in Quantum algorithms the oracle must be reversible



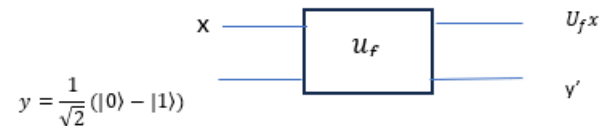
U_f can be seen as a unitary operator which performs.

$$U_f|x\rangle|y\rangle = |x\rangle|y \otimes f(x)\rangle$$

(Qskit 2020)

Our oracle will take in n input bits and output 1 bit. This will be either 1 if this is the element we are looking for or 0 if this is not the element, we are looking for.

Using this we can construct our phase oracle which is given by.



....THE ORACLE

This can be written as

$$(-1)^{f(x)}|x\rangle|y\rangle$$

(Qiskit, 2020)

Since this is independent of y we can rewrite the above equation. U_f is what we refer to as our phase oracle. It performs a mapping of

$$u_f|x\rangle = -1^{f(x)}|x\rangle$$

(Qiskit, 2020)

....GROVERS ALGORITHM

Remember our oracle.

The goal is to find our element w in our list

$U_f|x\rangle = -1^{f(x)}|x\rangle$ we can apply it so that we map $U_f: |w\rangle \rightarrow -|w\rangle$ and $|x\rangle \rightarrow |x\rangle$ ($\forall x \neq w$)

This can be written as

$$U_f = \mathbb{1} - 2|w\rangle\langle w|$$

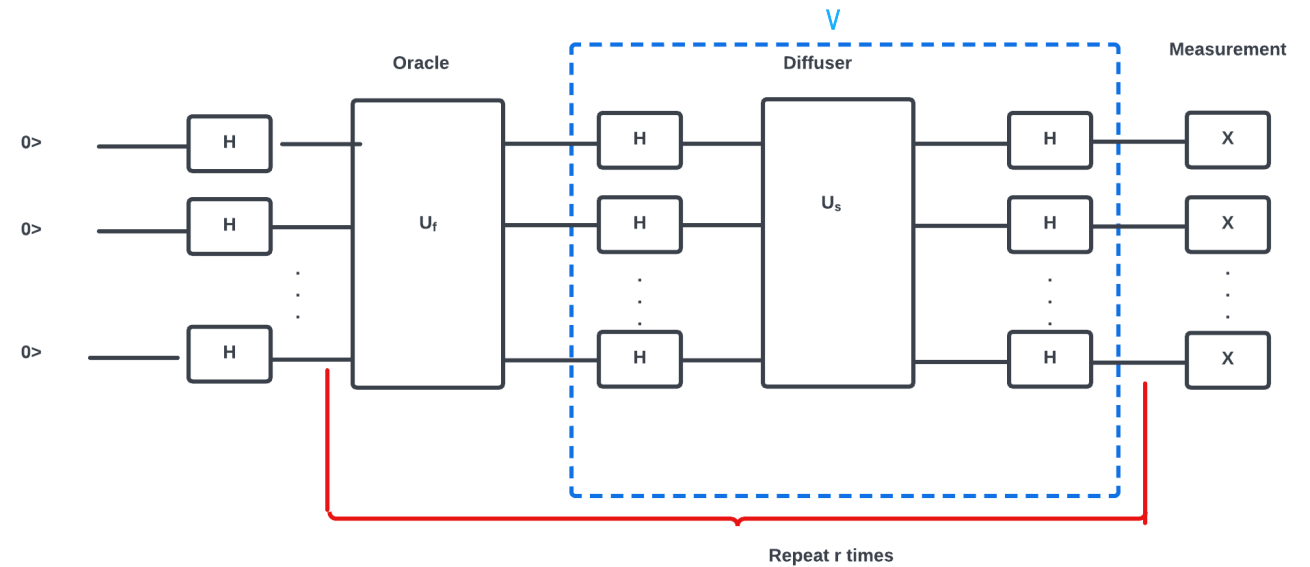
(Qiskit, 2020)

We can also define the other function $U_{f0}: |0\rangle^{\otimes n} \rightarrow |0\rangle^{\otimes n}$ and $|x\rangle \rightarrow -|x\rangle$ ($\forall x \neq 000$)

From this we can write $U_{f0} = 2|0\rangle\langle 0|^{\otimes n} - \mathbb{1}$

(Qiskit, 2020)

THE QUANTUM CIRCUIT FOR GROVER'S ALGORITHM



DERIVED FROM QISKIT 2020*

....GROVERS ALGORITHM

The state v can be given as

$$v = 2|s\rangle\langle s| - \mathbb{1}$$

S is the superposition state

(Qiskit, 2020)

What Grovers algorithm does is to carry out $(v \cdot U_f)^r$ on the state $|s\rangle$

Both the oracle and the operator $2|s\rangle\langle s| - \mathbb{1}$ reflect about $|s\rangle$ The combination of the two is a rotation hence we can define

$$|s\rangle = \cos\frac{\theta}{2}|s'\rangle + \sin\frac{\theta}{2}|w\rangle$$

HOW IT WORKS

Step 1

We start with uniform superposition $|s\rangle$ which is constructed by applying Hadamard gates $|s\rangle = H^{\otimes n}|0\rangle$

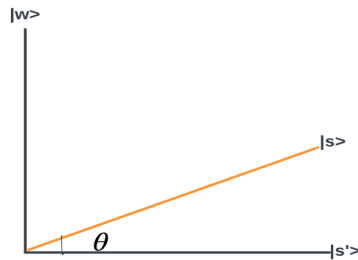
The aim is to find our solution w with our function $f(x) = \begin{cases} 1 & \text{if } x=w \\ 0 & \text{elsewhere} \end{cases}$

Step 2

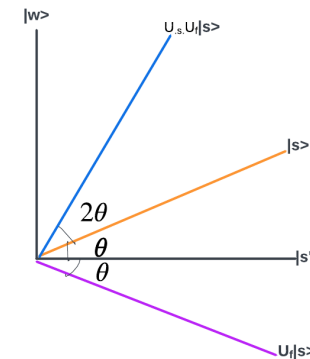
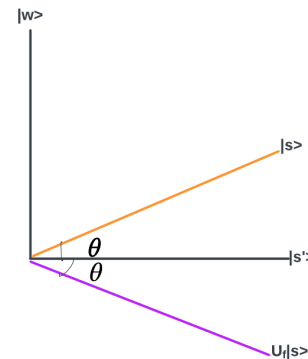
Apply the Oracle reflection $U_f = 1 - 2|w\rangle\langle w|$ to the state $|s'\rangle$. This corresponds to a reflection of the state $|s'\rangle$. This will lead to the lowering of the average amplitude as the state becomes negative.

Step 3

Apply additional reflection about the state $|s\rangle$, $U_s = 2|s\rangle\langle s| - I$. This completes the transformation. Rotates the initial state the initial state $|s\rangle$ close to our search goal. U_s action can be described as the reflection around the mean. Reflection at $|s\rangle$



Repeat Steps 2 and 3 basically until the state is close as possible to the winner. This can be defined $(U_s \cdot U_f)^{r \text{ times}}$ as on the state $|s\rangle$. Corresponds to a rotation by the angle 2θ



HOW MANY ITERATIONS?

$\sqrt{N}$ times... This is because the amplitude of $w\rangle$ grows linearly with the number of applications. To calculate the number of rotations we need to know the sample space and the number of solutions we are looking for. This can be given by the equation.

$$r = \left\lceil \frac{\pi}{4} \sqrt{\frac{N}{M}} \right\rceil \text{ if } N \text{ is large } \frac{\pi}{4} \sqrt{N} \text{ which comes to } O(\sqrt{N})$$

SOLVING THE SUDOKU PUZZLE AS DISCUSSED IN THE QISKIT TEXTBOOK

Step 1

v0	v1
v2	v3

We define the conditions in this case.

- in which case the conditions are $v0 \neq v1$ and $v2 \neq v3$ and $v0 \neq v2$ and $v1 \neq v3$ basically no row or column should have the same value twice. To make it simpler to work with the above conditions are converted into a list of clauses.
- `clause_list = [[0,1],[0,2],[1,3],[2,3]]` # Columns we need to make sure the rows/columns have different values

Step 2

Initialize the registers

- To do this we need 4 registers-:
- ✓ One register for our Sudoku variables (v0,v1,v2,v3).
- ✓ One register for the clauses defined above.
- ✓ One qubit where we will store the output to check the output circuit

The Oracle

The Oracle in this case uses two auxiliary registers in the state

$$U_w|x\rangle|0\rangle|-\rangle = \begin{cases} |x\rangle|0\rangle|-\rangle & \text{for } x \neq w \\ -|x\rangle|0\rangle|-\rangle & \text{for } x = w \end{cases}$$

- The result is that the output bit will be flipped if all clauses are satisfied. The oracle also returns the state back to $|000\rangle$ after computation.

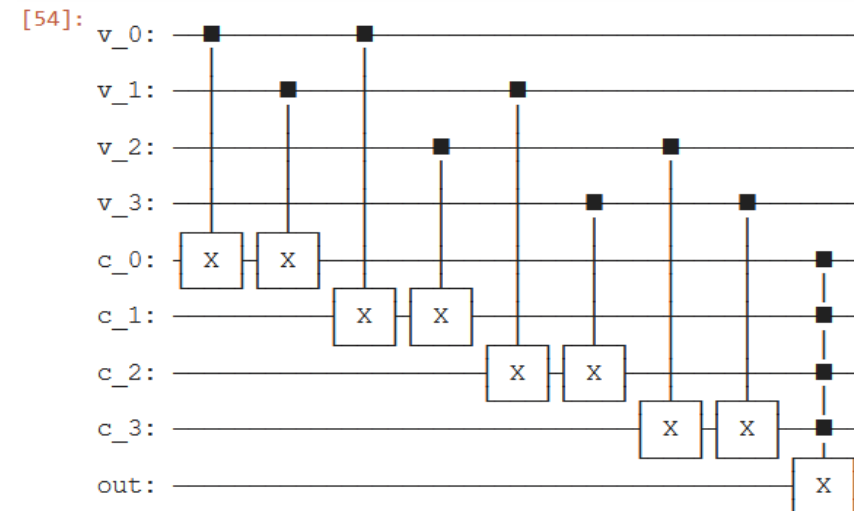
SOLVING THE SUDOKU PUZZLE AS DISCUSSED IN THE QISKIT TEXTBOOK

Step 3

- Define the Diffuser
- The purpose of the diffuser is to initialize the qubits into superposition state.

Step 4

- We add all the steps to our quantum circuit. Both the Oracle and the diffuser. We and repeat this 2 times and simulate the algorithm



The output will only be 1 if c_0 , c_1 , c_2 , c_3 is all 1. This defines our solution

SUMMARY

In this presentation we have looked at an example of quantum algorithm that is proven to perform faster than the classical algorithm. We have also illustrated how the Grover's algorithm can be used to solve a 2x2 sudoku puzzle where our solution is more efficient than the brute force attack using classical computer.

As quantum technology evolves and future designs will have to consider reducing the number of qubits needed and circuits for more efficient quantum algorithms



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